ACE LINER

Line following robot

What is ACE Liner?

ACE Liner is the line-tracing robot in the ACE lineup. It holds a special place in my heart because line tracing was the very first category I competed in when starting out in robotics. While the original version relied on a 5-channel line sensor, Version 2 upgrades to an 11-channel array alongside major structural overhauls. To maximize speed and agility, I redesigned the chassis to sit much lower to the ground and slimmed its thickness down from 5 mm to 3 mm to keep it lightweight.

Why ACE?

ACE started as a joke on my surname, ACEna, but it stands for “keep acing.” It’s the brand behind every robot in the lineup.

What was the Version 1?

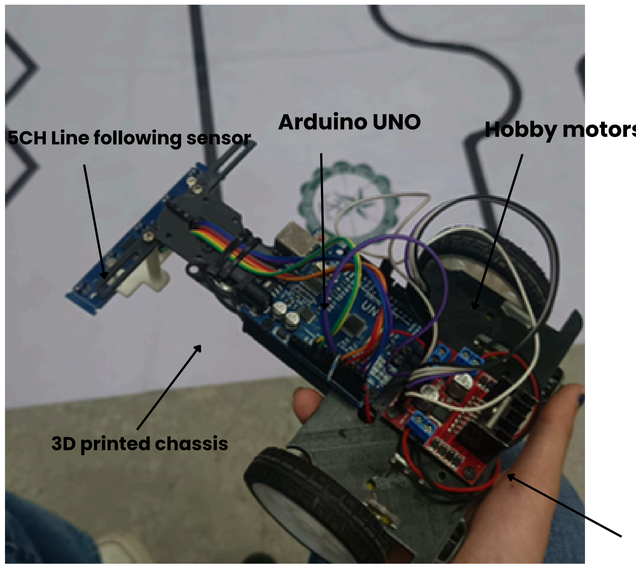
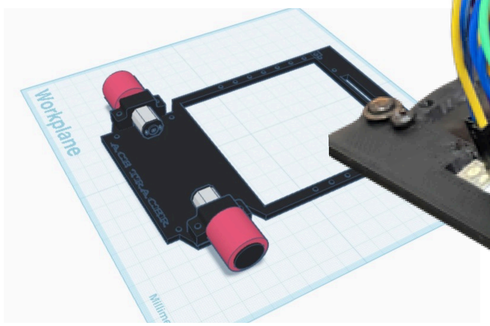


Figure 1. First ever line-following robot

This is the line-following robot my friend and I built for the National Robotics Competition (NRC) 2025. The 5-channel sensor was fine on the beginner track. On the intermediate track it lost the line at intersections, ran slow, and wiggled. We didn't place, but we learned a lot, and V2 was built to fix those problems.

This was modeled in TinkerCAD.



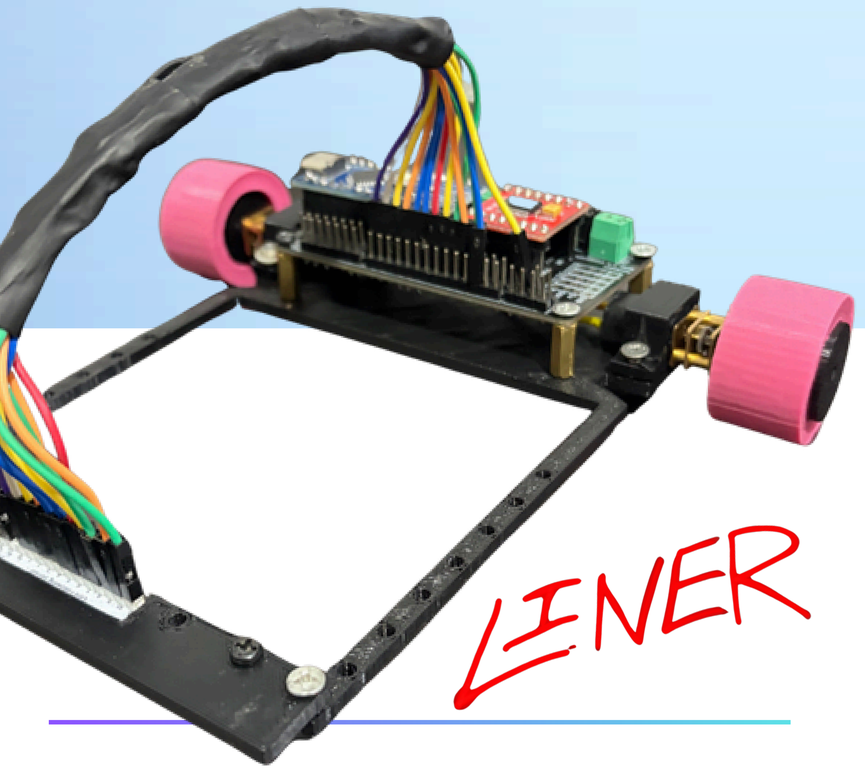


Figure 2. 3D Model figure of version 2

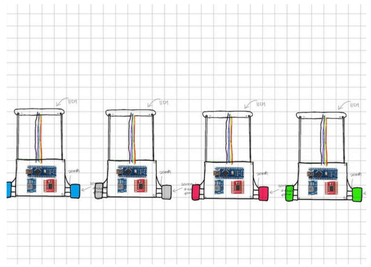
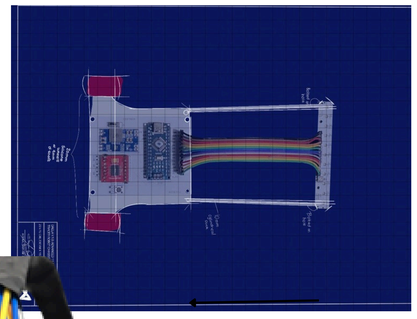


Figure 3 and 4. The decision-making process

I couldn't decide on wheel colors, so I drew several options and compared them on the chassis. I picked pink because I like it, because it's rare in competitions, and because it matches the red lights on my sensors.



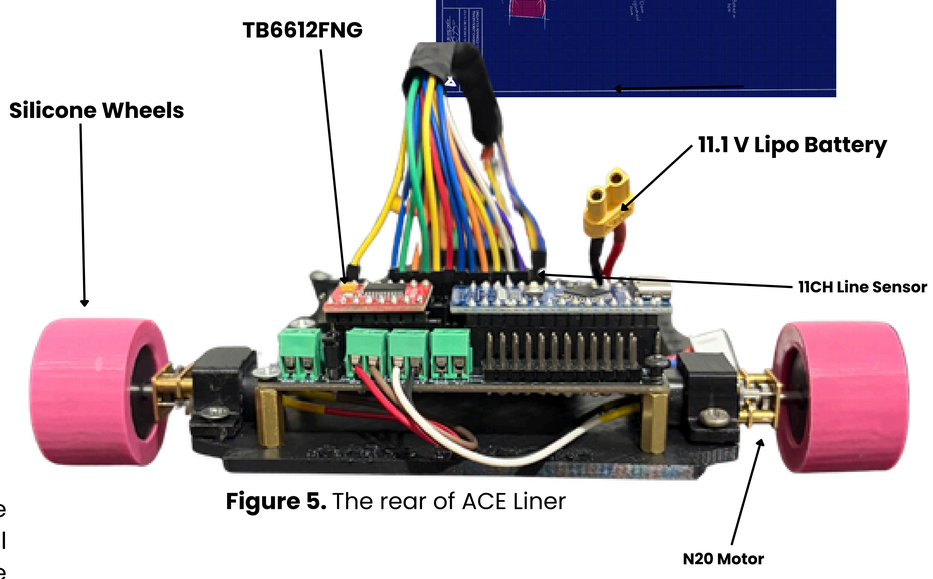


Figure 5. The rear of ACE Liner

The parts and the upgrade

Version 2 features a 100x160 mm 3D-printed chassis. The original yellow hobby motors were replaced with N20 motors running at 20,000 RPM, significantly reducing overall weight while boosting top speed. Powering the system is a 3S 11.1V LiPo battery managed by a TB6612FNG motor driver, bringing the total weight of the robot down to an agile 127g

Where it all started

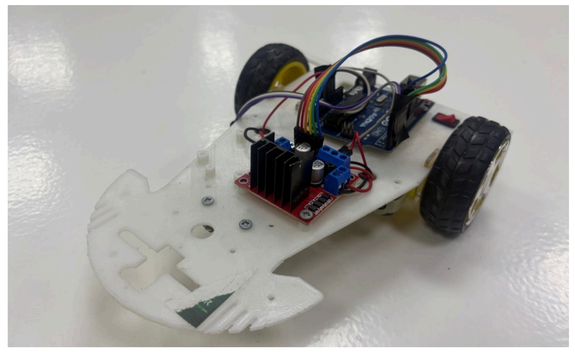


Figure 6. The Erovoutika Line Following Kit

I saw the chassis the first time I entered the robotics laboratory and was immediately curious about how to tinker with it.

ACE LINER: VERSION 1 TO VERSION 2

	Version 1	Version 2
Line Sensor	5-Channel IR Sensor	11-Channel IR Sensor
Motor	Yellow Hobby Motors	N20 - 20,000 RPM
Chassis	5mm thick	3mm, lower
Main Problem	Lost line at intersections, wiggly movement, and slow.	Fixed Version 1's problems and now lighter in weight

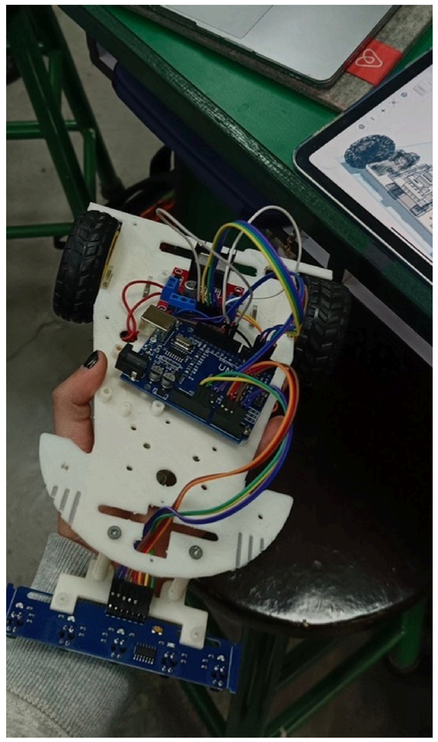


Figure 1. The draft before the Version 1

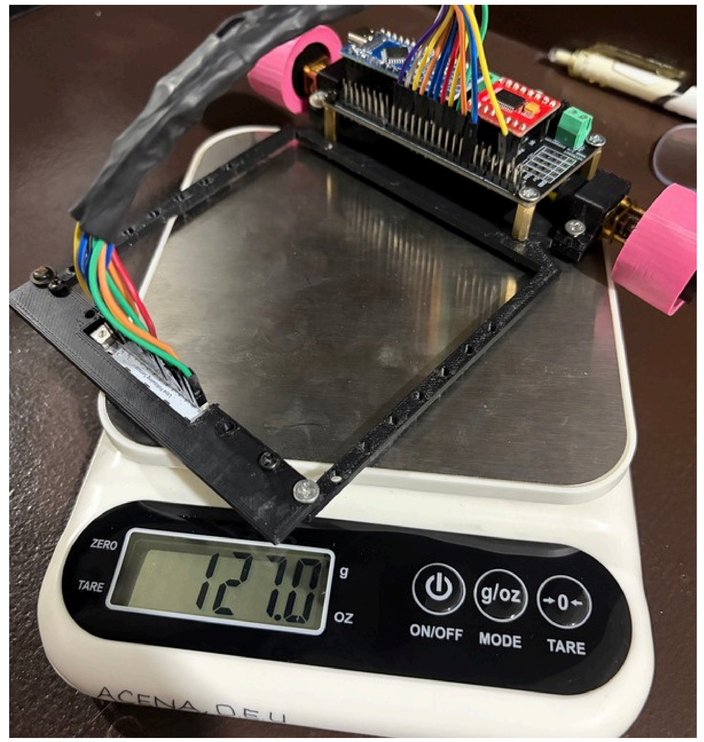


Figure 2. Weighing the Liner

What I learned: Five sensor channels were fine for the beginner track but not for intersections. A finer sensor array, lighter motors, and a thinner chassis were my answer to a slow, wiggly V1.